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The added value of extending the EQ-5D-5L with an itching item for the assessment of health-related quality of life of burn patients: an explorative study

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ABSTRACT

Introduction: Health-related quality of life (HRQL) is an important outcome in burn care and research. An advantage of a generic HRQL instrument, like the EQ-5D, is that it enables comparison of outcomes with other conditions and the general population. However, the downside is that it does not include burn specific domains, like scar issues or itching. Adding extra items to a generic instrument might overcome this issue. This study explored the potential and added value of extending the EQ-5D-5L with a burn-specific item, using a itching item as an example.
Methods: The EQ-5D-5L and the Patient and Observer Scar Assessment Scale (POSAS) was completed by adult patients 5–7 years after injury. A separate POSAS itching item was used to study the added value of an itching item for the EQ-5D-5L. The EQ-5D-5L + Itching was created by adding the POSAS itching item to the EQ-5D-5L. Five psychometric properties were compared between EQ-5D-5L and EQ-5D-5L + Itching: distribution (e.g. ceiling), informativity cf. Shannon's indices, convergent validity, dimension dependency, and explanatory power respectively.
Results: A total of 243 patients were included, of whom 49% reported any itching on the POSAS. Adding an itching item to the EQ-5D-5L decreased the ceiling effect, and resulted in increased absolute informativity (H' = 4.76 vs. H' = 3.64) and relative informativity (J' = 0.34 vs. J' = 0.31). The extra itching item decreased the convergent validity (Spearman's rank correlation coefficient = −0.51 vs. −0.59). Mutual dependency of dimensions existed, showing that all other items were dominant over the itching item. Adding the itching item to the standard EQ-5D-5L barely improved explanatory power (49.3% vs. 49.0%).
Conclusions: PThe present study showed adding a burn-specific item to the EQ-5D-5L is possible and has potential. However, 5 to 7 years after injury, adding an itching item to the EQ-5D-5L provides little additional information; the gain in terms of added value is relatively small. Apart from instances where itching information is specifically needed, a strong case is not present for adding an itching item to the EQ-5D-5L for long-term (>5 yr after burns) HRQL

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assessment in burn patients. In early time periods after burn, the added value might be greater and we recommend exploring this potential in future studies, ideally on multiple timepoints after burn.

1. Introduction

Health-related quality of life (HRQL) is an important outcome in burn care and research [1,2]. This outcome is increasingly studied to understand, qualify and quantify the impact of burn injuries and to assess patient values. HRQL is a self-reported outcome that reflects a patient's perception of the impact of disease, injury or any event on a patient’s physical, psychological and social wellbeing [3].

Instruments to measure HRQL of burn patients can be generic (applicable to patients with any health condition) or burn-specific (applicable to patients with burn injuries). Burn-specific instruments are sensitive to the typical effects of burn injuries [4], where generic instruments enable comparison among different health conditions, and usually have population standards. There is an ongoing discussion on what instrument is best to assess HRQL of burn patients [5].

A generic instrument might be favorable as such an instrument allows comparison with the general population and across different conditions and diseases. Besides, it enables burden of disease and cost-effectiveness estimations which are increasingly used for resource allocation and evaluation of care [6,7]. However, it might not cover the full HRQL of burn patients as it does not take into account burn specific complaints, like scar issues, itching, disfigurement, self-esteem, working ability, fatigue, personal relations, and sexual problems/dysfunction [8,9]. To overcome this issue, burn-specific items might be developed for generic instruments. A generic HRQL instrument that is increasingly used within the field of burns is the EQ-5D instrument [1]. The EQ-5D takes a few minutes to fill out, is cognitively simple, and is available in many languages [10]. The EQ-5D enables the use of so-called ‘bolt-on’ dimensions to cover lack of comprehensiveness [11]. A ‘bolt-on’ is a specific dimension (like cognition, vision) that covers a specific health problem or dysfunction that has not yet been covered by the original instrument [11]. Testing the potential gains of a burn-specific bolt-on addresses the question whether EQ-5D suffices for standard outcome measurement in burn-patients.

Skin and scar related problems are frequently reported by burn patients, and are of key relevance for perceived health [12,13]. Specific symptoms thereof include pain and itching [14,15]. Itching is not explicitly defined by one of the five existing EQ-5D items, however it might be captured in the EQ-5D-5L usual activities or pain/discomfort items. An EQ-5D bolt-on item on itching could provide additional important information to assess or monitor HRQL in burn patients. The aim of our study was to explore the potential and added value of extending the EQ-5D-5L with a burn-specific item, using an itching item as an example.

2. Methods

2.1. *Participants*

Adult burn patients (≥18 years old) with a hospital stay of ≥1 day or who had surgical treatment in a Dutch burn center between August 2011 and September 2012 were selected from the Dutch Burn Repository registry [16]. This sample was extended with patients with severe burns (>20% total body surface area (TBSA) burned in patients ≤50 years old; >10% TBSA burned in adults >50 years; or TBSA full thickness >5% [17]) admitted between January 2010 and March 2013. Eligible patients had no cognitive impairments and had sufficient knowledge of the Dutch language. These patients were invited to participate via a postal invitation, including an information letter, an informed consent form and the survey. Invitations were sent between March 2017 and March 2018. Patients who did not respond after 3 weeks received a phone call (or a postal reminder when telephone details were missing). Their potential participation was then discussed. The study was performed according to the principles of the Declaration of Helsinki, registered at the Netherlands Trial Register (NTR6407), and approved by the Ethics Committee (registration number NL59981).

2.2. *Measures*

Patient and clinical characteristics were extracted from the Dutch Burn Repository R3 [18]. Patient characteristics included age and sex, and clinical characteristics included percentage total body surface area (%TBSA) burned, % full-thickness burns, etiology, number of surgeries, length of hospital stay, and time since burn. The survey consisted of two existing questionnaires: the EQ-5D-5L and the patient scale of the Patient and Observer Scar Assessment Scale (POSAS).

The EQ-5D-5L consists of five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression [19]. In the 5L version, the response options are: no problems, slight problems, moderate problems, severe problems and extreme problems/unable to. Based on the responses to these five dimensions, a health profile (assigning an ordinal number to each level on each dimension, e.g. 13214), a total equally weighted summary score (level sum score: range 5–25), and a utility score (ranging from 0 (for a health state considered as bad as being dead) to 1 (full health)) were calculated based on the Dutch value set [20]. The utility score can also have a negative value for health states that are considered worse than death [19]. The participants also rated their health on a visual analogue scale (EQ VAS), ranging from 0 (“worst imaginable health”) to 100 (“best imaginable health”) [10].

The patient scale of the POSAS 2.0 includes six items covering pain, itching, color, thickness, relief and pliability of a scar [21,22]. The one separate item on itching was used in this study. Itching was scored on a 10-point scale ranging from 1 (no itch) to 10 (extreme itch). Participants completed the patient part of the POSAS for their most severe scar. We transformed the itching item as experimental 5L-bolt-on to the EQ-5D by recoding the response: 1 = 1 (no itching); 2–3 = 2; 4–5 = 3; 6–7 = 4; 8–10 = 5, assuming 1 to represent no problems and 8-10 to represent extreme problems.

2.3. Data analyses

Burn patients who completed all items of the EQ-5D-5L, the EQ VAS and the POSAS itching item were included in this study. A non-response analysis was performed comparing the characteristics of responders and non-responders. Differences were assessed with Mann Whitney U tests for continuous variables and chi-square tests for categorical variables. Descriptive statistics were used to assess characteristics and outcomes of the participants.

The separate POSAS itching item was used to study the added value of an itching item for the EQ-5D-5L. When comparing the EQ-5D with the EQ-5D + Itching, the latter consists of the EQ-5D plus the recoded POSAS itching item. Distributional effects of the EQ-5D-5L and EQ-5D-5L + Itching were studied. First the number of unique profiles was calculated. Then, the ceiling of the EQ-5D-5L and EQ-5D-5L + Itching was determined by comparing the proportion of full health profiles (i.e. 11111 for the EQ-5D-5L and 111111 for the EQ-5D-5L + Itching). The higher the proportion of perfect health profiles, the higher the ceiling. Also, the effect of itching on the EQ VAS was investigated by studying the mean EQ VAS for each of the five recoded levels of itching, and any outcomes on the EQ-5D.

Subsequently, the informativity of the EQ-5D-5L with and without the itching item was assessed by determining the Shannon index (H’) and the Shannon Evenness index (J’) [23,24]. These indices provide information on the ability of the EQ-5D-5L(+Itching) to measure diversity and to discriminate in a population [25]. The Shannon index (H’) was calculated as: $H' = -\sum^{C}_{i=1} p_i \, {}^2\log p_i$, with p_i being the proportion of patients with one specific health profile, and C being the total number of possible health profiles. The higher the value of the Shannon index (H’), the more information is actually captured by the EQ-5D-5L or EQ-5D-5L + Itching. H'_{max} is the same metric, but theoretically assuming all possible health profiles of the classification are actually observed, and have the same proportion; this defines the theoretical maximal informativity of H given a classification. H'_{max} is, by definition, ${}^2\log C$. Note that classification A can have a lower theoretical maximum H'_{max} than B, and still have a higher observed H’ than B, due to more efficient use of the response options in terms of informativity (as demonstrated for some comparable dimensions of the EQ-5D-3L versus the more refined HUI2 or HUI3 [23]).

The Shannon Evenness index (J’) was calculated as: $J' = H'/H'_{max}$, If H' is higher with the itching item added, then more information is empirically captured; while J’ reflects relative informativity expressing the use of the classifications given their potential. Both H’ and J’ need to be taken into account for a full interpretation of informativity when comparing classifications [26]. All calculations were separately done for the EQ-5D-5L and the EQ-5D-5L + Itching. The maximal number of unique classification profiles is 3,125 categories (5*5*5*5*5) for the EQ-5D-5L; and 15,625 categories for the EQ-5D-5L + Itching. As the number of patients in our study (n = 245) was lower than the maximal number of unique profiles in either EQ-5D-5 L or EQ-5D-5L + Itching, H' could not reach the maximum a priori.

To compare the convergent validity of the EQ-5D-5L and the EQ-5D-5L + Itching, the Spearman’s rank correlation coefficient of the EQ VAS with both measures (using the level sum score) was computed. Next, we analyzed in detail the mutual relations among all dimension-specific information components (the five dimensions of the EQ-5D-5L and the itching item) to discover redundancy and dependency patterns. Dimension dependency was checked by inspection of dominance relations in cross tables of the EQ-5D-5L + Itching items. Profiles with a severe or extreme problem in one dimension (L4: severe problem or L5: extreme problem) and no problem (L1: no problem) in another dimension were studied [27]. We compared the number and percentage of L1-L4/L5 versus L4/L5-L1 contrasts for combinations of dimensions that included the extra itching item.

Lastly, we empirically tested the added explanatory power of itching information through regression analyses, taking the EQ VAS score as external reference. We predicted the EQ VAS score from the EQ-5D-5L dimensions with or without the itching item, and without and with socio-demographic factors.

First we applied univariate analyses to test the relation between the EQ-5D-5L + Itching domains and the EQ VAS. The levels ‘slight problems’, ‘moderate problems’, ‘severe problems’ and ‘extreme problems’ were used for the prediction of the EQ VAS; the ‘no problems’ level was used as the reference category. Then multivariate analyses (backward selection) were carried out with all the EQ-5D-5L dimensions, both with and without the itching item in the model, and with and without age, sex and comorbidity. The significance level was universally set at $p < 0.05$. IBM SPSS Statistics 23 was used for all analyses.

3. Results

3.1. Characteristics

A total of 257 patients of the 517 eligible patients were willing to participate, of whom 243 (47%) completed the EQ-5D-5L, the EQ VAS, and the POSAS. Responders were older ($p < 0.01$) and more often females ($p = 0.04$) than non-responders (Appendix 1). The characteristics of participants are outlined in Table 1. Participants were most often male (62%) and the mean age was 48 years (SD 17). Median %TBSA burned was 6% (IQR 2-12) and median length of stay was 11 days (IQR 2–24). Five to seven years after burn, participants had a mean utility score of 0.87 (SD 0.18). The mean EQ VAS was 81.6 (SD 15.7) (Table 1). Most problems were reported on the EQ-5D pain/discomfort dimension; 39% of the participants reported at least slight problems. Itching (POSAS itching score ≥ 2) was reported by almost half of the participants (n = 118; 49%).

Table 1 – Demographic characteristics of study population.

	Total sample (n = 243)
Sex: male, n (%)	151 (62.1%)
Age at survey (M, SD)	47.5 (16.7)
Age at burn (M, SD)	41.9 (16.7)
%TBSA (Median, IQR)	6.0 (2.0-12.0)
Length of hospital stay (Median, IQR)	11.0 (2.0-24.0)
Number of surgeries (Median, IQR)	1.0 (0.0-1.0)
Aetiology, n (%)	
Flame	139 (57.2%)
Scald	46 (19.1%)
Other	56 (23.2%)
Comorbidity, n (%)	
No	182 (74.9%)
One	43 (17.7%)
More than one	18 (7.4%)
Time since burn (years) (M, SD)	5.6 (0.5)
EQ-5D-5L scores	
Utility score (M, SD)	0.87 (0.18)
EQ VAS (M, SD)	81.6 (15.7)
Mobility (% with problems)	15.6%
Self-care (% with problems)	7.8%
Usual activities (% with problems)	23.5%
Pain/discomfort (% with problems)	39.1%
Anxiety/depression (% with problems)	31.3%
POSAS Patient Scale scores	
Itching (M, SD)	2.6 (2.2)
% patients with itching (POSAS itching score $\geq$ 2)	48.6%

M, SD = mean, standard deviation; IQR = interquartile range.

Table 2 – EQ VAS score for each of the five recoded itching items.

Itching score	Number of patients	Mean EQ VAS (95% CI)
Itching = 1	125	83.3 (80.7–85.9)
Itching = 2	58	83.0 (79.2–86.8)
Itching = 3	24	75.7 (67.1–84.3)
Itching = 4	26	77.6 (70.5–84.7)
Itching = 5	10	77.5 (69.2–85.9)

3.2. *Distributional effects - ceiling, unique profiles*

In total, 113 of the 243 participants (46.5%) reported a perfect health status on the EQ-5D-5L, and 78 (32.1%) reported a perfect health status based on the EQ-5D-5L + Itching. The ceiling was thus higher for the EQ-5D-5L compared to the EQ-5D-5L + Itching. The 35 participants that had a perfect health status based on the EQ-5D-5L, but not based on the EQ-5D-5L + Itching were mainly males (77.1%). These patients had a mean age of 45 years (SD 16) at survey and a median % TBSA of 4% (IQR 2–9). Their median length of hospital stay was 7 days (IQR 2–16). They reported an EQ VAS of 88.8 (SD 12.7).

The number of different health profiles in the whole sample of participants was 59 out of 3,125 (1.8%) for the EQ-5D-5L, and 86 out of 15,625 (0.6%) for the EQ-5D-5L + Itching. The EQ-5D-5L + Itching seemed thus to better differentiate between patients compared to the EQ-5D-5L.

Table 2 shows the mean EQ VAS score for each level of itching and any EQ-5D outcome. In general, the mean EQ VAS score decreased when itching problems were worse. Mean EQ VAS outcomes were comparable for participants with no itching problems and slight itching problems (83.3 and 83.0, respectively), and for participants experiencing moderate to extreme itching problems (75.7–77.6). Itching problems seemed thus to influence the overall experienced health status of patients.

3.3. *Informativity of the EQ-5D-5L with and without an extra itching item*

The Shannon Index (H') was 4.76 for the EQ-5D-5L + Itching and 3.64 for the EQ-5D-5L. This indicates that more information was captured by the extended measure compared to the EQ-5D-5L. The Shannon Evenness index (J') of the EQ-5D-5L + Itching was higher than that of the EQ-5D-5L (0.34 vs. 0.31), implying that informativity increased even when correcting for its maximum potential.

3.4. *Convergent validity*

The Spearman's rank correlation coefficient (between EQ-5D-5L level sum score and EQ VAS) was −0.587, and −0.507 between the EQ-5D-5L + Itching (level sum score) and the EQ VAS. The convergent validity was thus higher for the EQ-5D-5L, meaning that the EQ-5D-5L without extra itching item was more strongly related to the EQ VAS compared to the EQ-5D-5L + Itching.

3.5. *Dimension dependency*

In Table 3 the combinations of extreme values of the EQ-5D-5L + Itching (no versus severe/extreme problems) are presented in absolute numbers and expressed as the percentage of the total EQ-5D-5L + Itching profiles. We found that severe/extreme problems on itching and no problems on other items were relatively common (2.5%–13.2%), especially for no problems on mobility (11.9%) and self-care (13.2%). Conversely, no problems on itching and severe/extreme problems on other items were relatively uncommon (0–1.2%). This indicates that the other items were dominant over the itching item.

3.6. *Explanatory power of the EQ-5D-5L with and without an itching item*

None of the levels of the itching item were statistically significantly associated with the average EQ VAS score (Appendix 2). The multivariate regression analyses results are displayed in Table 4. In total, 49.0% of the variance of the EQ VAS was explained by the five EQ-5D-5L dimensions. Adding an extra itching item (49.3%) did not provide much additional explanatory power. Especially the anxiety/depression dimension explained a large part of the variance; the explained variance without the anxiety/depression dimension is 34.5%. Self-care did not seem to explain any variance, as the explained variance of the model with and without the item self-care was 49.3%. The final multivariate model, that also included age and sex, included no levels of the extra itching

Table 3 – Combinations of extreme values of the EQ-5D-5 L + Itching (no versus severe/extreme problems), in absolute numbers and expressed as the percentage of the total EQ-5D-5 L + Itching profiles (n = 243).

		No problems					
		Mobility	Self-care	Usual activities	Pain/ discomfort	Anxiety/ depression	Itching
Severe/extreme problems	Mobility		1 (0.4%)	1 (0.4%)	2 (0.8%)	3 (1.2%)	3 (1.2%)
	Self-care	0		0	0	0	0
	Usual activities	0	0		0	0	0
	Pain/ discomfort	0	3 (1.2%)	1 (0.4%)		1 (0.4%)	3 (1.2%)
	Anxiety/ depression	6 (2.5%)	6 (2.5%)	5 (2.1%)	3 (1.2%)		2 (0.8%)
	Itching	29 (11.9%)	32 (13.2%)	17 (7.0%)	6 (2.5%)	19 (7.8%)	

Table 4 – Explanatory power of multivariate models for the EQ VAS that include EQ-5D-5L dimensions with and without an extra itching item added.

Selection of EQ-5D-5 L + Itching items	R^2	F-value	p-value
MO + SC + UA + PD + AD	0.490	12.7	<0.001
SC + UA + PD + AD + IT	0.475	12.0	<0.001
MO + UA + PD + AD + IT	0.493	11.4	<0.001
MO + SC + PD + AD + IT	0.463	11.4	<0.001
MO + SC + UA + AD + IT	0.463	10.7	<0.001
MO + SC + UA + PD + IT	0.345	6.7	<0.001
MO + SC + UA + PD + AD + IT	0.493	10.2	<0.001

Note. MO = mobility, SC = selfcare, UA = usual activities, PD = pain/discomfort, AD = anxiety/depression, IT = itching.

item, nor any levels of the self-care item. This final model explained 50.6% of the variance of the EQ VAS (Table 5).

4. Discussion

This study showed that adding a burn-specific item to the EQ-5D-5L is possible and has potential. The addition of an itching item provides some additional information, however, the added value is small 5 to 7 years after burns. BB By definition, the ceiling of the EQ-5D-5L decreased due to the addition of the itching item, and more information is captured by the EQ-5D-5L + Itching. Moreover, adding an itching item to the EQ-5D-5L increased absolute as well as relative informativity. The convergent validity, with the EQ VAS as reference, was lower in the EQ-5D-5L + Itching. The pattern of severe/extreme itching problems and no problems on other dimensions, occurs more frequently than the reverse: severe/extreme problems on other dimensions and no itching problems, indicating that the other items are dominant over the itching item. In addition, adding the extra itching item provided little extra explanatory power to the EQ-5D-5L.

Earlier studies displayed a need for an itching item in other fields. Within the field of psoriasis, the bolt-on dimension ‘skin irritation (operationalized using an itching item)’ was added to the EQ-5D-5L as explanatory factor analysis indicated that this dimension emerged as a separate important factor [28]. Another study developed a model for mapping an itching item to the EQ-5D-3L based on a general population sample, because itching has a high impact on HRQL of patients with systemic or skin diseases [29,30]. Also within burn patients, itching has been shown to impact HRQL [31]. However, results from our study showed that adding an itching item to the EQ-5D-5L in a burn sample did only provide small additional explanatory power.

Table 5 – Multivariate model for the EQ VAS, including EQ-5D-5 L dimensions, extra itching item and sociodemographic factors.

EQ-5D-5L dimension	Unstandardized B	p-value
Constant	93.282	<0.001
Age	−0.141	0.002
Sex	3.134	0.041
Mobility level 2	−6.948	0.003
Usual activities level 2	−7.235	0.025
Anxiety/depression level 2	−9.209	<0.001
Usual activities level 3	−10.653	0.020
Anxiety/depression level 3	−8.893	<0.001
Pain/discomfort level 4	−24.698	<0.001
Anxiety/depression level 4	−27.895	<0.001
Anxiety/depression level 5	−37.104	<0.001
F value	23.7	<0.001
R-square	0.506	

Besides, the results on convergent validity indicated that the itching item did not have a positive impact as a bolt-on item to the EQ-5D-5L, which may indicate that itching is already covered by the other EQ-5D-5L dimensions, e.g. in the discomfort aspect of the pain/discomfort dimension, which was observed in an earlier study [32].

The informativity of the EQ-5D-3L has been studied before in trauma populations; Shannon indexes between 1.71 and 5.58 and Shannon Evenness indexes between 0.33 and 0.71 have been reported [33,34]. The Shannon index found in our study (3.63) was within this range, whereas the Shannon Evenness index was lower (0.31). This might be induced by the limited sample size, as Shannon Evenness is dependent on the total possible number of unique profiles, which is considerably higher than the current sample size, obstructing a fair estimate of relative informativity. Furthermore, the timing of the assessment might play a role, which was 5–7 years after injury in our study compared to 1 month and 2 years after injury in previous studies. A larger variation in outcomes might be expected shorter after injury, as well as in general trauma populations compared to burn injuries as a broader range of injuries with a wider variation in complaints and recovery patterns is included [34].

For future studies, we recommend to examine the added value of an extra itching item on HRQL of burn patients shorter after burns (e.g. within the first three months after sustaining burns) when itching is reported to be more severe and higher prevalent [14], especially since itching has been found to be associated with lower HRQL [31]. Also, the present study had a cross-sectional design. To further explore the added value of an extra itching item, investigating the change in itching problems within patients, measured with repeated measurements design, is insightful and would provide insight into the responsiveness of the EQ-5D-5L + Itching. In present study, we showed that adding an extra burn-specific item to the EQ-5D-5L is possible and has potential. We used an itching item as example, but one might think of other potential valuable items to extend the EQ-5D-5L to capture burn-specific complaints not included in the original EQ-5D items, for example skin/scar problems, work, body image, heat sensitivity, sexuality or fatigue [35,36].

This study had a number of strengths and limitations. A strength was the high prevalence of itching in the sample which allowed us to study the possibility and potential of extending the EQ-5D-5L with an itching item. Another strength was that both instruments, the EQ-5D-5L and the POSAS, were validated in the burn population [21,37]. A limitation was the relatively low variability in itching reported by the patients, only few patients reported severe itching, potentially due to the long-term follow-up data that we used. A second limitation was that itching was measured with the POSAS, which required mapping of the POSAS 10-level scale to the EQ-5D 5-level scale. This mapping was arbitrary and may have invoked autocorrelation in several of the psychometric analysis. Due to lack of refinement, ceiling effect, and overestimation of reported problems, measurement bias is assumed to be higher in a 5-level scale [38]. This may have led to less problems reported on the 10-level scale compared to when it would have been assessed at a 5-level scale. A third limitation was the different timeframes in the instruction of the instruments (POSAS assesses scar quality in the last weeks) and the assessment of itching in the scar versus the whole body.

5. Conclusions

Present study showed that adding a burn-specific item to the EQ-5D-5L is possible and has potential. However, 5 to 7 years after injury, adding an itching item to the EQ-5D-5L provides only some additional information; the gain in terms of added value is small. Apart from instances were itching information is specifically needed, there is not a strong case for adding an itching item to the EQ-5D-5L for long-term (>5 yr after burns) HRQL assessment in burn patients. Shorter after burns the added value might be greater and we recommend to explore this potential in future studies, ideally on multiple timepoints after burn.

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Conflicts of Interest Statement

All authors whose names are listed immediately below have no financial and personal relationships with other people or organisations that could inappropriately influence (bias) their work. Examples of potential conflicts of interest include employment, consultancies, stock ownership, honoraria, paid expert testimony, patent applications/registrations, and grants or other funding.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.burns.2020.08.015.

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